

1. Screenshot when you perform step#3

Test your cluster with nginx as followed:

- Deploy an nginx on your cluster using the deployment file from <https://k8s.io/examples/application/deployment.yaml>
- Use kubectl to describe the deployment

```
(base) PS C:\Users\USER\Desktop\Software_def\act4> kubectl describe pod nginx-deployment
Name:          nginx-deployment-bf744486c-659wc
Namespace:     default
Priority:       0
Service Account: default
Node:          docker-desktop/192.168.65.3
Start Time:    Tue, 01 Sep 2026 19:14:19 +0700
Labels:        app=nginx
               pod-template-hash=bf744486c
Annotations:   <none>
Status:        Running
IP:            10.1.0.10
IPs:           IP: 10.1.0.10
Controlled By: ReplicaSet/nginx-deployment-bf744486c
Containers:
  nginx:
    Container ID:  docker://c015870274a64dd2a87810bfbc24cc5ead8bece710db1787a218064135febd0
    Image:         nginx:1.14.2
    Image ID:      docker-pullable://nginx@sha256:f7988fb6c02e0ce69257d9bd9cf37ae20a60f1df7563c3a2a6abe24160306b8d
    Port:          80/TCP
    Host Port:     0/TCP
    State:         Running
      Started:     Tue, 01 Sep 2026 19:14:30 +0700
    Ready:         True
    Restart Count: 0
    Environment:   <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-hjcdk (ro)
Conditions:
  Type                               Status
  PodReadyToStartContainers          True
  Initialized                         True
  Ready                              True
  ContainersReady                    True
  PodScheduled                       True
```

2. The deployment file that you create in step#4b and ingress file in step#4d
 - b. Create a deployment file of a pod with service of the following containers:
 - i. todo (release-2.1)
 - ii. redis

Note: you can use a docker-hub image: natawut/todo-service:release-2.1 and you must ensure that environment for REDIS_HOST

```
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: todo-service
5  spec:
6    selector:
7      matchLabels:
8        app: todo-service
9    template:
10     metadata:
11       labels:
12         app: todo-service
13     spec:
14       containers:
15         - name: todo-service
16           image: natawut/todo-service:release-2.1
17           env:
18             - name: REDIS_HOST
19               value: "localhost"
20           resources:
21             limits:
22               memory: "128Mi"
23               cpu: "500m"
24           ports:
25             - containerPort: 8000
26         - name: redis
27           image: redis
28           ports:
29             - containerPort: 6379
30
```

d. Create ingress configuration file, apply the file, and show the ingress information

```
(base) PS C:\Users\USER\Desktop\Software_def\act4> kubectl apply -f https://raw.githubusercontent.com/kubernetes/ingress-nginx/controller-v1.12.0/deploy/static/provider/cloud/deploy.yaml
namespace/ingress-nginx unchanged
serviceaccount/ingress-nginx unchanged
serviceaccount/ingress-nginx-admission unchanged
role.rbac.authorization.k8s.io/ingress-nginx unchanged
role.rbac.authorization.k8s.io/ingress-nginx-admission unchanged
clusterrole.rbac.authorization.k8s.io/ingress-nginx unchanged
clusterrole.rbac.authorization.k8s.io/ingress-nginx-admission unchanged
rolebinding.rbac.authorization.k8s.io/ingress-nginx unchanged
rolebinding.rbac.authorization.k8s.io/ingress-nginx-admission unchanged
clusterrolebinding.rbac.authorization.k8s.io/ingress-nginx unchanged
clusterrolebinding.rbac.authorization.k8s.io/ingress-nginx-admission unchanged
configmap/ingress-nginx-controller configured
service/ingress-nginx-controller unchanged
service/ingress-nginx-controller-admission unchanged
deployment.apps/ingress-nginx-controller configured
job.batch/ingress-nginx-admission-create unchanged
job.batch/ingress-nginx-admission-patch unchanged
ingressclass.networking.k8s.io/nginx created
validatingwebhookconfiguration.admissionregistration.k8s.io/ingress-nginx-admission configured
```

```
(base) PS C:\Users\USER\Desktop\Software_def\act4> kubectl apply -f .\ingress.yml
ingress.networking.k8s.io/todo-ingress created
```

```
⦿ (base) PS C:\Users\USER\Desktop\Software_def\act4> kubectl describe nginx-ingress-controller
error: the server doesn't have a resource type "nginx-ingress-controller"
● (base) PS C:\Users\USER\Desktop\Software_def\act4> kubectl describe services nginx-ingress-controller
Name:          nginx-ingress-controller
Namespace:     default
Labels:        app.kubernetes.io/instance=nginx-ingress
               app.kubernetes.io/managed-by=Helm
               app.kubernetes.io/name=nginx-ingress
               app.kubernetes.io/version=5.5.4
               helm.sh/chart=nginx-ingress-2.6.4
Annotations:   meta.helm.sh/release-name: nginx-ingress
               meta.helm.sh/release-namespace: default
Selector:      app.kubernetes.io/instance=nginx-ingress,app.kubernetes.io/name=nginx-ingress
Type:          LoadBalancer
IP Family Policy: SingleStack
IP Families:   IPv4
IP:            10.105.169.243
IPs:           10.105.169.243
LoadBalancer Ingress: localhost
Port:          http 80/TCP
TargetPort:    80/TCP
NodePort:      http 30669/TCP
Endpoints:     10.1.0.13:80
Port:          https 443/TCP
TargetPort:    443/TCP
NodePort:      https 31778/TCP
Endpoints:     10.1.0.13:443
Session Affinity: None
External Traffic Policy: Local
Internal Traffic Policy: Cluster
HealthCheck NodePort: 30324
Events:        <none>
⦿ (base) PS C:\Users\USER\Desktop\Software_def\act4>
```

3.

4-c

```
(base) PS C:\Users\USER\Desktop\Software_def\act4> kubectl get services
NAME                                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
kubernetes                         ClusterIP           10.96.0.1       <none>           443/TCP          64m
nginx-ingress-controller           LoadBalancer       10.105.169.243  localhost        80:30669/TCP,443:31778/TCP 13m
(base) PS C:\Users\USER\Desktop\Software_def\act4> kubectl apply -f .\deploy_service.yml
deployment.apps/todo-service created
Error from server (NotFound): endpoints "todo-service" not found
(base) PS C:\Users\USER\Desktop\Software_def\act4> kubectl expose deployment/todo-service
service/todo-service exposed
(base) PS C:\Users\USER\Desktop\Software_def\act4> kubectl get endpoints todo-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 Endpointslice
NAME                ENDPOINTS                                AGE
todo-service        10.1.0.14:6379,10.1.0.14:8000           3s
```

4-e

```
(base) PS C:\Users\USER\Desktop\Software_def\act4> curl.exe --% -X POST http://localhost/ -H "Content-Type: application/json" -d '{"title":"\
"todo-1","detail":"the first todo","duedate":"2024-10-13 11:00:00","tags":[],"completed":false}'
{"id":0}
(base) PS C:\Users\USER\Desktop\Software_def\act4> curl localhost

Security Warning: Script Execution Risk
Invoke-WebRequest parses the content of the web page. Script code in the web page might be run when the page is parsed.
RECOMMENDED ACTION:
Use the -UseBasicParsing switch to avoid script code execution.

Do you want to continue?

[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "N");
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "N"); y

StatusCode      : 200
StatusDescription : OK
Content          : [{"id":0,"title":"todo-1","detail":"the first todo","completed":false,"duedate":"2024-10-13T11:00:00","tags":[]}]
RawContent       : HTTP/1.1 200 OK
                  Connection: keep-alive
                  Content-Length: 113
                  Content-Type: application/json
                  Date: Tue, 01 Sep 2026 13:10:39 GMT
                  Server: nginx/1.31.3

Forms           : {}
Headers         : [{"Connection, keep-alive"}, [{"Content-Length, 113"}, [{"Content-Type, application/json"}, [{"Date, Tue, 01 Sep 2026 13:10:39
                  GMT}]]]]
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 113
```

```
(base) PS C:\Users\USER\Desktop\Software_def\act4> .\activity4-CP_windows_amd64.exe
2026/09/08 13:29:58 Current Timestamp: 2026-09-08T13:29:58+07:00
👉 Enter Kubernetes namespace (leave blank for 'default'):
👉 Enter domain (leave blank for 'http://localhost'):
2026/09/08 13:29:59 ✅ / Namespace exists and can use kubectl command.
2026/09/08 13:30:00 ✅ / All Kubernetes resources are up and running.
2026/09/08 13:30:00 ✅ / Ingress resource exists in the namespace.
2026/09/08 13:30:00 ✅ / URL http://localhost returns 200.
2026/09/08 13:30:00 ✅ / Todo is up and running at http://localhost
2026/09/08 13:30:00 ✅ / Todo service is inaccessible at http://localhost:8000
2026/09/08 13:30:00 ✅ / Redis service is inaccessible at http://localhost:6379
2026/09/08 13:30:00 ✅ / POST request successful
2026/09/08 13:30:00 ✅ / POST request to http://localhost was successful.
2026/09/08 13:30:00 ✅ / GET request shows result from previous POST request to http://localhost.
2026/09/08 13:30:00 Result: true
2026/09/08 13:30:00 🎉 Looks good! Please enter your StudentID and Full name below
👉 StudentID: 6630199021
👉 Full Name (TH): เภาสาณ กสิธร
2026/09/08 13:30:09 🎉 Successfully submitted. Your lucky number is 21467360763559426
2026/09/08 13:30:09 🎉 Congratulations! You have completed the activity 🎉
2026/09/08 13:30:09 ⚠️ Don't forget! you still need to submit your assignment via MyCourseVille ⚠️
(base) PS C:\Users\USER\Desktop\Software_def\act4>
```